

Service providers are looking to have solutions from companies which can address the challenges in rolling out networks in the rural areas

—A. Sethu Raman, executive director of Huawei Technologies India



"Rural market has different requirements, price phenomena and challenges compared to the urban market and thus needs different

products," explains Keskar. So one needs to understand the requirements, work with product companies to enable technical specifications and price target that are suitable for rural market needs, and develop relevant cost-effective applications. Most of the technology firms are instrumental in providing various solutions that service providers use to create product designs especially targeted for the rural segment.

Looking for ideas? Take a peek below to know how some of the top players in the field are playing the game!

Creating customised protocols

Freescalcaters to power needs of rural India

Rural markets need maintenance-free vehicles. Freescale, together with local vendors of two-wheeler manufacturers, has developed low-cost cluster/ignition and body electronics solutions that improve efficiency/emission and fuel consumption of motorcycles. These solutions add to features and safety of the vehicles.

Power consumption is a big concern in rural areas. Freescale has worked closely with leading UPS/inverter manufacturers to develop high-efficiency products based on its DSP controller technology.

The threat of global warming and climate change due to excessive use of fossil fuels increases the urgency of finding environmentally benign ways of generating energy. Freescale has worked on developing solutions for solar-based inverters that drive water pump for farmers. It has also developed LED lantern solutions that can replace kerosene lamps from the rural market by offering a cost-effective and green solution for lighting.

In the networking and telecom space, Freescale has worked with local telecom companies to help launch a low-power GSM base station based on Freescale Power PC and StarCore DSP. It can be powered by solar energy as well and can provide mobile connectivity to rural markets.

Empowering finance

mChck makes rural transaction simpler

mChck is focusing on the vast 'unbanked' masses of people to facilitate mobile-based transactions at the micro level, as over 30 per cent of the new mobile subscriptions every month come from villages.

To implement its mobile-based secure authentication and payment platform for transactions, mChck tied up with rural microfinance institution Grameen Koota for the pilot launch.

Grameen Koota serves over 160,000 members throughout Karnataka. Through its model, mChck enabled Grameen Koota microfinance borrowers to receive loans and make loans repayments and merchant and bill payments in the local villages easily and sufficiently through mobile phones.

NXP's solution for branchless rural banking

Having pioneered the branchless banking solutions, NXP initiated successful pilot projects in the villages of Andhra Pradesh that reached out to almost 45,000 people with leading banks such as State Bank of India, Union Bank, Axis Bank, Andhra Bank, State Bank of Hyderabad and Andhra Pradesh Grameena Vikas Bank.

Now, NXP is working with banks in Karnataka to take this initiative to the rural areas of Bengaluru. Under the aegis of the State Bank committee of Karnataka, various banks such as Syndicate Bank, Canara Bank and Corporation Bank are part of this initiative.

The solution is based on NXP's near-field communication (NFC) and contactless smart card technologies utilising NXP chipsets. Typically, a smart card stores the identity of the customer, including such details as name, address, photograph, fingerprint templates and relevant details of the savings or loan accounts held by the issuing bank, in a small chip embedded in the card and provides for authentication of the consumer through the biometric fingerprint. An NFC-enabled mobile phone acts as a branch of the bank by storing the entire database of customers in the village and neighbouring areas within the phone's memory, protected by a very high-security PKI chip built into the NFC subsystem on the phone. The system is fast, secure and reliable, empowering banks to slash costs per transaction and provide multiple services to the customers.

The solution has the potential of providing full-featured banking services—like cash deposits, cash withdrawals, utility payments, money transfers, micro-insurance and cashless payments—to rural consumers.